

# Module/Pathway/Taxon Review: Pentose Phosphate Pathway (ppu00030) in *Pseudomonas putida* KT2440

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**Target taxon:** *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Target bucket:** KEGG `ppu00030` — "Pentose phosphate pathway" (resolved local bucket with 19 primary genes; 36 candidate genes; module area central\_carbon) **Prepared for:** manual module satisfiability and gene-annotation curation

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## 1. Executive Summary

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The classical **pentose phosphate pathway (PPP)** is **fully satisfiable** in *Pseudomonas putida* KT2440. Every canonical step of both the oxidative branch (glucose-6-phosphate dehydrogenase, 6-phosphogluconolactonase, decarboxylating 6-phosphogluconate dehydrogenase) and the non-oxidative branch (ribose-5-phosphate isomerase, ribulose-5-phosphate 3-epimerase, transketolase, transaldolase), plus the PRPP entry point (ribose-phosphate pyrophosphokinase), is encoded in the genome with no reaction gaps. The satisfiability conclusion is **high-confidence at the pathway level** because pathway operation is supported by direct  $^{13}\text{C}$  metabolic flux analysis ( $^{13}\text{C}$ -MFA) and in-vitro enzyme assays in KT2440, even though the individual gene assignments are only homology-inferred (UniProt protein-existence level 3).

The most important curation issue is **not** a gap but a **boundary/over-annotation problem**. The KEGG `ppu00030` bucket is over-broad: it bundles together three biochemically distinct processes — (i) the PPP proper, (ii) the **Entner-Doudoroff (ED) pathway**, and (iii) **peripheral periplasmic glucose/gluconate oxidation** — plus PRPP synthesis and several broadly-mapped isomerases/mutases. Of the 36 candidate genes, only ~10 encode PPP-proper enzymes (plus 2 for PRPP entry). The remaining ~15–24 candidates are ED enzymes (`edd`, `eda`), peripheral-oxidation enzymes (`gcd`, `gnuK`, `kguK`, gluconate-2-dehydrogenase subunits, `adhB`), gluconeogenic/EDEMP enzymes (`fba`, `fbp`), or likely over-propagated annotations from unrelated pathways (`trxB`, `ptxD`, `phnN`, `ttuD`, `rbsK`, `pgm`, `algC`, `cpsG`).

The key species-specific context is that **KT2440 lacks a functional Embden-Meyerhof-Parnas (EMP) pathway** (no phosphofructokinase) and catabolizes glucose almost exclusively through the ED route, operating a cyclic **EDEMP** architecture that recycles ~10% of triose phosphates back to hexose phosphates for NADPH overproduction. As a consequence, the oxidative PPP carries only **minor catabolic flux** and the non-oxidative branch is **flux-limited** (biosynthetic/anabolic role rather than a glycolytic conduit). Curation recommendation: mark all core PPP steps **covered**, keep ED as a **separate module**, and flag the peripheral-oxidation, PRPP, gluconeogenic, and redox genes as **not\_expected\_in\_target\_taxon as PPP steps / likely over-annotation**. The bucket itself should be marked **module\_needs\_revision**.

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## 2. Target-Organism Pathway Definition

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**What the PPP is (scope to keep):** The pentose phosphate pathway is the set of reactions that (a) oxidize glucose-6-phosphate (G6P) to ribulose-5-phosphate (Ru5P) with generation of NADPH and CO<sub>2</sub> (oxidative branch), and (b) interconvert pentose phosphates and other sugar phosphates (Ru5P  $\rightleftharpoons$  ribose-5-phosphate/xylulose-5-phosphate; C3–C7 sugar-phosphate shuffling by transketolase/transaldolase) to supply R5P for nucleotide biosynthesis and E4P for aromatic amino-acid biosynthesis, and to interface with glycolysis/gluconeogenesis at F6P and GAP (non-oxidative branch).

**Neighboring processes to keep separate (do NOT satisfy PPP steps with these):**

| Neighboring process                                                                    | Genes co-mapped in ppu00030                                                                                                                                                                                           | Correct home                                                                         |
|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Entner-Doudoroff pathway</b> (6PG → KDPG → pyruvate + GAP)                          | <code>edd</code> (PP_1010), <code>eda</code> (PP_1024)                                                                                                                                                                | KEGG module M00008 / map ppu00030 co-display, but a distinct pathway                 |
| <b>Periplasmic glucose/gluconate oxidation</b> (glucose → gluconate → 2-ketogluconate) | <code>gcd</code> (PP_1444), <code>gnuK</code> (PP_3416), <code>kguK</code> (PP_3378), <code>PP_3382/3383/3384</code> (gluconate-2-DH), <code>adhB</code> (PP_3623), <code>ghrB</code> (PP_1261), <code>PP_1661</code> | Peripheral glucose catabolism (upstream feeder reactions)                            |
| <b>PRPP biosynthesis</b>                                                               | <code>prs</code> (PP_0722), <code>PP_2744</code> , <code>phnN</code> (PP_4469)                                                                                                                                        | Nucleotide-precursor branch (M00005); <code>phnN</code> is phosphonate metabolism    |
| <b>Gluconeogenesis / EDEMP recycling</b>                                               | <code>fba</code> (PP_4960), <code>fbp</code> (PP_5040)                                                                                                                                                                | Glycolysis/gluconeogenesis (map ppu00010)                                            |
| <b>Redox / other pathways</b> (likely over-propagation)                                | <code>trxB</code> (PP_0786), <code>ptxD</code> (PP_3376), <code>ttuD</code> (PP_4300), <code>rbsK</code> (PP_2458), <code>pgm</code> / <code>algC</code> / <code>cpsG</code>                                          | Thioredoxin, phosphonate, hydroxypyruvate, ribose salvage, nucleotide-sugar/alginate |

**Alternate names / database definitions:** KEGG map ppu00030 explicitly notes that the map "also shows the Entner-Doudoroff pathway where 6-P-gluconate is dehydrated and then cleaved into pyruvate and glyceraldehyde-3P," which is why ED enzymes appear in the bucket. The PPP is decomposed by KEGG into modules **M00006** (oxidative phase, G6P→Ru5P), **M00007** (non-oxidative phase, F6P→R5P), and **M00005** (PRPP biosynthesis), and the map additionally co-displays **M00008** (Entner-Doudoroff) and **M00004** (PP cycle). Curators should treat "PPP" as M00006 + M00007 (+ optionally M00005 as an entry point), and explicitly exclude M00008.

### 3. Expected Step Model

The canonical PPP step model and its coverage status in KT2440:

## OXIDATIVE BRANCH (M00006)

G6P --[G6P dehydrogenase, EC 1.1.1.49]--> 6-phosphogluconolactone [COVERED: zwf/zwfA/zwfB]  
 6-PGL --[6-phosphogluconolactonase, EC 3.1.1.31]--> 6-phosphogluconate [COVERED: pgl (+ppgL?)]  
 6-PG --[6-PG dehydrogenase (decarb.), EC 1.1.1.44]--> Ru5P + CO<sub>2</sub> + NADPH [COVERED: gntZ]

## NON-OXIDATIVE BRANCH (M00007)

Ru5P --[R5P isomerase A, EC 5.3.1.6]--> R5P [COVERED: rpiA]  
 Ru5P --[Ru5P 3-epimerase, EC 5.1.3.1]--> Xu5P [COVERED: rpe]  
 Xu5P + R5P --[transketolase, EC 2.2.1.1]--> S7P + GAP [COVERED: tktA]  
 S7P + GAP --[transaldolase, EC 2.2.1.2]--> F6P + E4P [COVERED: tal, flux-lim]  
 Xu5P + E4P --[transketolase, EC 2.2.1.1]--> F6P + GAP [COVERED: tktA]

## PRPP ENTRY (M00005)

R5P --[ribose-P pyrophosphokinase, EC 2.7.6.1]--> PRPP [COVERED: prs (+PP\_2744?)]

## LINKER

G6P <--[G6P isomerase, EC 5.3.1.9]--> F6P [COVERED: pgi1/pgi2]

**Every canonical step maps to at least one KO-assigned candidate gene, with no missing reaction.** The oxidative branch has paralog redundancy at the first step (three G6PD genes) and a single copy of the decarboxylating dehydrogenase ([gntZ](#)). The non-oxidative branch is single-copy at every step. KEGG module→gene links confirm exactly one gene per canonical non-oxidative step (M00007 = tktA, tal, rpe, rpiA), and place both [pgl](#) and the candidate second lactonase [ppgL](#) within the oxidative module M00006.

## 4. Candidate Genes and Evidence

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### 4.1 PPP-proper core genes (high confidence — mark covered)

| Step (EC / KO)                                          | Gene(s)                                      | Locus                           | Evidence                                    | Curation caveat                                                                                                                                                                                                                                   |
|---------------------------------------------------------|----------------------------------------------|---------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| G6P dehydrogenase (1.1.1.49 / K00036)                   | <b>zwf</b> ,<br><b>zwfA</b> ,<br><b>zwfB</b> | PP_5351,<br>PP_1022,<br>PP_4042 | Homology (UniProt PE3); three real paralogs | Paralog redundancy; <b>zwf</b> primary. In <i>P. bharatica</i> CSV86, ZwfA is a dual cofactor-specific isozyme predominant in glucose metabolism (PMID: <a href="#">36354357</a> ) — suggests functional differentiation worth checking in KT2440 |
| 6-phosphogluconolactonase (3.1.1.31 / K01057)           | <b>pgl</b>                                   | PP_1023                         | Homology (PE3)                              | <b>ppgL</b> (PP_2021, K07404) is a candidate second lactonase — <b>candidate_uncertain</b> ; annotation score 1/5, no recommended enzyme name                                                                                                     |
| 6-PG dehydrogenase, decarboxylating (1.1.1.44 / K00033) | <b>gntZ</b>                                  | PP_4043                         | Homology (PE3); single copy                 | Low catabolic flux; <b>gntZ</b> deletion improves muconate production (PMID: <a href="#">31931111</a> ) — confirms it is a real, active branch                                                                                                    |
| R5P isomerase A (5.3.1.6 / K01807)                      | <b>rpiA</b>                                  | PP_5150                         | Homology (PE3)                              | Single copy                                                                                                                                                                                                                                       |
| Ru5P 3-epimerase (5.1.3.1 / K01783)                     | <b>rpe</b>                                   | PP_0415                         | Homology (PE3)                              | Local metadata assigns primary bucket ppu00040 (over-broad); functionally PPP                                                                                                                                                                     |
| Transketolase (2.2.1.1 / K00615)                        | <b>tktA</b>                                  | PP_4965                         | Homology (PE3)                              | Single copy                                                                                                                                                                                                                                       |
| Transaldolase (2.2.1.2 / K00616)                        | <b>tal</b>                                   | PP_2168                         | Homology (PE3)                              | <b>Flux-limited</b> — engineering studies implant exogenous transketolase/transaldolase to reinforce PPP (PMID: <a href="#">38531855</a> )                                                                                                        |
|                                                         | <b>prs</b>                                   | PP_0722                         |                                             | <b>PP_2744</b> is a paralog, <b>candidate_uncertain</b> (PE4)                                                                                                                                                                                     |

| Step (EC / KO)                       | Gene(s)                      | Locus               | Evidence       | Curation caveat                                          |
|--------------------------------------|------------------------------|---------------------|----------------|----------------------------------------------------------|
| PRPP synthase (2.7.6.1 / K00948)     |                              |                     | Homology (PE3) | "Predicted," score 2, no gene symbol                     |
| G6P/F6P isomerase (5.3.1.9 / K01810) | <b>pgi1</b> ,<br><b>pgi2</b> | PP_1808,<br>PP_4701 | Homology       | Linker isomerase shared with glycolysis/ gluconeogenesis |

## 4.2 Entner-Doudoroff genes (present in genome; keep as SEPARATE module)

**edd** (PP\_1010, K01690, phosphogluconate dehydratase) and **eda** (PP\_1024, K01625, KDPG aldolase) are the core ED enzymes. In KT2440 these carry the **dominant** glycolytic flux (the strain routes glucose to 6-phosphogluconate and cleaves it via Edd/Eda to pyruvate and glyceraldehyde-3-phosphate), but they are **not PPP steps** — they belong to KEGG module M00008 and should be curated under an ED module, not counted toward PPP satisfiability.

## 4.3 Peripheral glucose/gluconate oxidation (upstream feeders; not PPP)

**gcd** (K00117, quinoprotein glucose dehydrogenase), **gnuK** (K00851, gluconokinase), **kguK** (K11441, 2-ketogluconokinase), **PP\_3382/3383/3384** (gluconate-2-dehydrogenase subunits, K06150/1/2), **adhB** (K06150), **ghrB** (K00090), and **PP\_1661** (K19813) constitute the periplasmic oxidation network that converts glucose → gluconate → 2-ketogluconate before carbon enters central metabolism as 6-phosphogluconate. These are biochemically upstream feeder reactions, **not** PPP steps.

## 4.4 Likely over-propagated / other-pathway annotations (mark not\_expected as PPP)

**trxB** (PP\_0786, K22345, thioredoxin reductase — redox, not PPP), **ptxD** (PP\_3376, phosphonate dehydrogenase), **phnN** (PP\_4469, K05774, phosphonate metabolism), **ttuD** (PP\_4300, hydroxypyruvate reductase), **rbsK** (PP\_2458, ribokinase — ribose salvage), **pgm** / **algC** / **cpsG** (phosphoglucomutase/phosphomannomutase — nucleotide-sugar/alginate), **PP\_3443** (gapN, K00131), and **fba** / **fbp** (gluconeogenesis/EDEMP). None of these should satisfy a canonical PPP step.

## 5. Gaps, Ambiguities, and Likely Over-Annotations

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**No reaction gaps.** Every canonical PPP step has an assigned gene. The oxidative and non-oxidative branches, plus PRPP entry and the G6P/F6P linker, are all covered.

### Key ambiguities:

1. **Zwf paralog assignment.** Three G6PD paralogs (`zwf`, `zwfA`, `zwfB`) is unusual. Which one carries oxidative-PPP flux vs. ED-feeding flux vs. oxidative-stress response is not resolved by homology alone. The *P. bharratica* CSV86 finding that ZwfA is a dual-cofactor (NAD<sup>+</sup>/NADP<sup>+</sup>) predominant isozyme ([PMID: 36354357](#)) is suggestive but is **weak-to-moderate species transfer** (different *Pseudomonas* species).
2. `ppgL` (**PP\_2021**) as second lactonase. UniProt annotation score 1/5, no recommended enzyme name → **candidate\_uncertain**. Do not count as a required step (`pgl` already covers it). KEGG places it in oxidative module M00006, so it is plausibly a genuine second lactonase but needs confirmation.
3. `PP_2744` as second PRPP synthase. UniProt PE4 "Predicted," no gene symbol → **candidate\_uncertain**.
4. **Evidence-level mismatch.** All core PPP gene annotations are **homology-inferred** (UniProt PE3), whereas pathway operation is **experimentally proven** by <sup>13</sup>C-MFA and enzyme assays. Satisfiability is therefore strong at the pathway level but individual gene-to-function mappings remain inferential.

**Over-annotation summary:** ~15 of the 36 candidates (ED, peripheral oxidation, PRPP, gluconeogenesis, redox) are not PPP-proper. The bucket bundles ≥3 distinct processes; this is the dominant curation problem.

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## 6. Module and GO-Curation Recommendations

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### Per-step status:



| Canonical step                                                 | Status                                                             | Gene(s)                                                                                                              |
|----------------------------------------------------------------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| G6P dehydrogenase                                              | <b>covered</b>                                                     | zwf (primary); zwfA, zwfB (paralogs)                                                                                 |
| 6-P-gluconolactonase                                           | <b>covered</b>                                                     | pgl; ppgL = candidate_uncertain                                                                                      |
| 6-P-gluconate DH (decarb.)                                     | <b>covered</b>                                                     | gntZ (single copy)                                                                                                   |
| R5P isomerase A                                                | <b>covered</b>                                                     | rpiA                                                                                                                 |
| Ru5P 3-epimerase                                               | <b>covered</b>                                                     | rpe                                                                                                                  |
| Transketolase                                                  | <b>covered</b>                                                     | tktA                                                                                                                 |
| Transaldolase                                                  | <b>covered</b> (flag flux-limited)                                 | tal                                                                                                                  |
| PRPP synthase                                                  | <b>covered</b>                                                     | prs; PP_2744 = candidate_uncertain                                                                                   |
| G6P/F6P isomerase (linker)                                     | <b>covered</b>                                                     | pgi1, pgi2                                                                                                           |
| Entner-Doudoroff (edd/eda)                                     | <b>separate module</b> (present, not a PPP step)                   | edd, eda                                                                                                             |
| Peripheral oxidation, PRPP-phosphonate, redox, gluconeogenesis | <b>not_expected_in_target_taxon as PPP steps / over-annotation</b> | gcd, gnuK, kguK, PP_3382-3384, adhB, ghrB, PP_1661, trxB, ptxD, phnN, ttuD, rbsK, pgm, algC, cpsG, fba, fbp, PP_3443 |

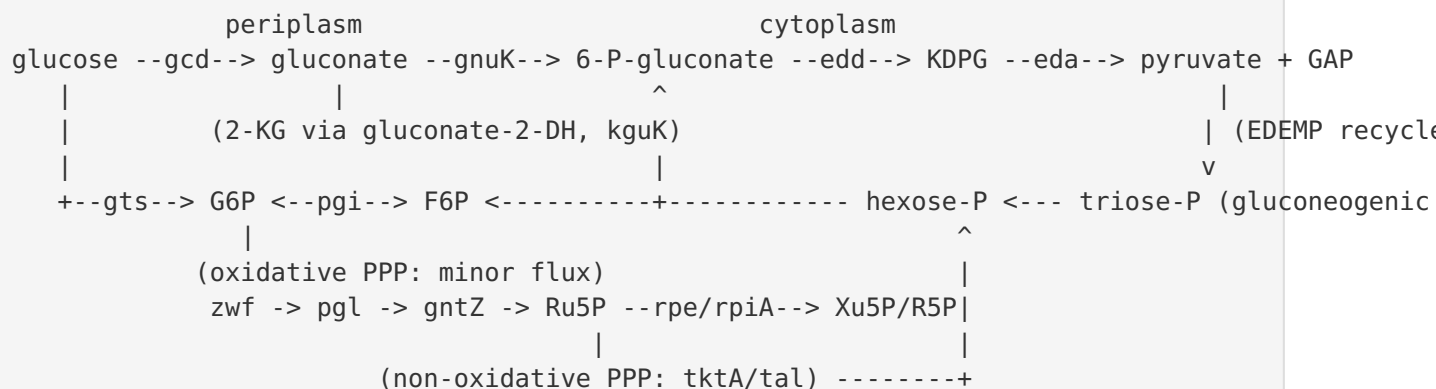
**Bucket-level recommendation:** mark `kegg:ppu00030` **module\_needs\_revision** — the generic KEGG boundary is wrong for curation purposes because it co-mingles PPP, ED, and peripheral oxidation. Recommend splitting into: (a) oxidative + non-oxidative PPP (M00006 + M00007), (b) a distinct ED module (M00008), and (c) a peripheral periplasmic glucose-oxidation module. No new GO terms appear necessary; standard PPP GO terms suffice, but the gene-to-term mappings for the ~15 over-annotated genes should be down-weighted or removed from the PPP context.

## 7. Genes to Promote to Full fetch-gene Review

1. **zwf / zwfA / zwfB (PP\_5351 / PP\_1022 / PP\_4042)** — resolve paralog roles (oxidative PPP vs. ED-feeding vs. stress); cofactor specificity.
2. **ppgL (PP\_2021)** — very low annotation confidence (score 1/5); confirm or reject as second 6-P-gluconolactonase.
3. **PP\_2744** — "Predicted" second PRPP synthase; confirm functionality vs. prs.
4. **gntZ (PP\_4043)** — single-copy decarboxylating dehydrogenase; deletion phenotype documented in muconate engineering; confirm it is the sole 6PGDH.
5. **tal (PP\_2168)** — flux-limiting transaldolase flagged by pentose-engineering studies; verify it is the only transaldolase.

## 8. Mechanistic Model / Interpretation

KT2440's central-carbon architecture reframes what "PPP satisfiability" means for this organism. Because the strain **lacks phosphofructokinase**, linear EMP glycolysis is nonfunctional, and glucose is instead funneled — largely through periplasmic oxidation to gluconate/2-ketogluconate — into **6-phosphogluconate**, which is then cleaved by the **Entner-Doudoroff** enzymes Edd/Eda to pyruvate and GAP:



In this **EDEMP cycle**, ~10% of triose phosphates are recycled back to hexose phosphates by a merged set of ED, gluconeogenic-EMP, and PPP reactions, tuned to overproduce NADPH. Consequently:

- The **oxidative PPP** (zwf→pgl→gntZ) is present and active but carries **minor catabolic flux**; its main role is NADPH supply, not glucose disposal.
- The **non-oxidative PPP** (rpe/rpiA/tktA/tal) functions primarily in an **anabolic/biosynthetic** direction (R5P for nucleotides, E4P for aromatics) and is **flux-limited** — engineering efforts must reinforce transketolase/transaldolase to route pentoses efficiently.
- The **ED pathway** is the true glycolytic workhorse and must be curated as a separate entity, even though KEGG co-displays it on the PPP map.

This is why the correct curation stance is "PPP fully present, but the bucket is over-broad and the pathway is flux-minor" rather than "PPP is the main glucose route."

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## 9. Evidence Base

| PMID                     | Paper                                                                                                                              | Role in this review                                                                                                                                                                                                 |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">26350459</a> | <i>P. putida</i> KT2440 metabolizes glucose through a cycle formed by enzymes of the ED, EMP, and PPP pathways (Nikel et al. 2015) | <b>Primary direct evidence.</b> <sup>13</sup> C-MFA + enzyme assays establish ED dominance, absent functional EMP, and the ED/EMP cycle recycling ~10% of triose-P; documents oxidative/non-oxidative PPP operation |
| <a href="#">17483213</a> | Convergent peripheral pathways catalyze initial glucose catabolism in <i>P. putida</i> (del Castillo et al. 2007)                  | Confirms three convergent peripheral pathways converging on 6-phosphogluconate, metabolized by Edd/Eda                                                                                                              |
| <a href="#">18245293</a> | A set of activators and repressors control peripheral glucose pathways in <i>P. putida</i>                                         | Regulatory architecture (HexR, GnuR, PtxS, GltR-2) of peripheral oxidation and ED genes — supports separating these from PPP                                                                                        |
| <a href="#">18493743</a> | Physiological states and energetic adaptation of <i>P. putida</i> mt-2 on glucose                                                  | Glucose → gluconate → 2-ketogluconate flux; periplasmic oxidation dominates over biosynthesis                                                                                                                       |
| <a href="#">31931111</a> | Engineering glucose metabolism for enhanced muconic acid production in <i>P. putida</i> KT2440                                     | <code>gcd</code> , <code>gntZ</code> , <code>hexR</code> deletions reshape flux — confirms <code>gntZ</code> and peripheral-oxidation genes are real and active                                                     |
| <a href="#">38531855</a> | Synthetically-primed adaptation of <i>P. putida</i> to D-xylose                                                                    | Implanting exogenous transketolase/transaldolase to reinforce PPP — supports "covered but flux-limited" for <code>tal</code> / <code>tktA</code>                                                                    |
| <a href="#">36354357</a> | ZwfA, a dual cofactor-specific isozyme, in <i>P. bharatica</i> CSV86                                                               | Suggests G6PD paralog functional differentiation; weak-to-moderate species transfer to KT2440                                                                                                                       |
| <a href="#">41141486</a> | Engineering mixed sugar metabolic channels                                                                                         | Context on pentose (xylose/arabinose) utilization and PPP interfacing                                                                                                                                               |
| <a href="#">30831266</a> | GC-MS-based analysis of <i>Pseudomonas</i>                                                                                         | Methodological / genus context                                                                                                                                                                                      |

### Verified supporting quotes (from stored abstracts):

- **PMID: 26350459:** *"The soil bacterium Pseudomonas putida KT2440 lacks a functional Embden-Meyerhof-Parnas (EMP) pathway, and glycolysis is known to proceed almost exclusively through the Entner-Doudoroff (ED) route."* — establishes ED dominance and absent functional EMP.
- **PMID: 26350459:** *"about 10% of the triose phosphates were found to be recycled back to form hexose phosphates. This set of reactions merges activities belonging to the ED, the EMP (operating in a gluconeogenic fashion), and the pentose phosphate pathways to form an unforeseen metabolic architecture (EDEMP cycle)"* — defines the EDEMP cycle intertwining PPP with ED and EMP.
- **PMID: 17483213:** *"glucose catabolism in Pseudomonas putida occurs through the simultaneous operation of three pathways that converge at the level of 6-phosphogluconate, which is metabolized by the Edd and Eda Entner/Doudoroff enzymes to central metabolites"* — confirms convergence on 6-phosphogluconate and ED exit.
- **PMID: 38531855:** *"We then enhance the pentose phosphate pathway by implanting exogenous transketolase and transaldolase..."* — supports native non-oxidative PPP being flux-limited.

## 10. Limitations and Knowledge Gaps

- **Gene-level annotations are homology-inferred.** All core PPP genes are UniProt protein-existence level 3 ("Inferred from homology"); `PP_2744` is level 4 ("Predicted") and `ppgL` has a 1/5 annotation score. Direct biochemical characterization of most individual KT2440 PPP enzymes is lacking — the strong evidence is at the **pathway** (flux) level, not the **gene** level.
- **Paralog roles unresolved.** The functional division of labor among the three `zwf` paralogs is not established from these data.
- **Second-copy uncertainty.** Whether `ppgL` and `PP_2744` are genuine functional second copies or spurious annotations is unresolved.
- **Species transfer.** The ZwfA cofactor-specificity insight comes from *P. bharatika*, not KT2440 — transfer is uncertain.
- **KEGG boundary artifact.** The `ppu00030` bucket's over-breadth is a database-definition artifact (KEGG deliberately co-displays ED on the PPP map); this is a curation-tooling issue, not a biological gap.

## 11. Proposed Follow-up Experiments / Actions

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1. **Promote to full fetch-gene review:** `zwf` / `zwfA` / `zwfB`, `ppgL`, `PP_2744`, `gntZ`, `tal` (see Section 7).
  2. **Split the bucket:** Create/curate three distinct modules — oxidative+non-oxidative PPP (M00006+M00007), Entner-Doudoroff (M00008), and peripheral periplasmic glucose oxidation — replacing the monolithic `ppu00030` mapping. Mark `kegg:ppu00030` **module\_needs\_revision**.
  3. **Down-weight over-annotations:** Remove or flag PPP-context GO/EC mappings for `trxB`, `ptxD`, `phnN`, `ttuD`, `rbsK`, `pgm`, `algC`, `cpsG`, `fba`, `fbp`, `PP_3443`.
  4. **Experimental resolution (expert questions):** (a) Single- and combinatorial `zwf` paralog knockouts with  $^{13}\text{C}$ -MFA to assign each paralog's flux role and cofactor preference; (b) enzymatic confirmation of `ppgL` lactonase activity and `PP_2744` PRPP-synthase activity; (c) transaldolase/transketolase activity assays to quantify the flux limitation flagged by xylose-engineering studies.
  5. **Verify single-copy assumptions:** Confirm `gntZ`, `rpiA`, `rpe`, `tktA`, `tal` are the sole genes for their reactions via genome-wide EC/KO scan.
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*Prepared as a species-aware module satisfiability and annotation-curation review. Satisfiability conclusions are high-confidence at the pathway level (direct  $^{13}\text{C}$ -MFA in KT2440) and moderate at the individual-gene level (homology-inferred). The dominant curation action is bucket revision, not gap-filling.*

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